FACULTY OF COMPUTING

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TITLE: BIG DATA PROCESSING IN DATA PROCESSING

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BIG DATA PROCESSING IN DATA ENGINEERING

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EXECUTIVE SUMMARY

This paper explains the growing importance of big data processing in modern data engineering, driven by the rapid increase in data generated from digital sources. The main five aspects of big data is volume, velocity, variety, veracity, and value. The study highlights why organizations rely on advanced big data processing techniques. However, organizations face critical challenges, including handling large-scale and diverse datasets, ensuring data quality, managing high costs, and maintaining data security and privacy. This paper also discusses the evolution of big data technologies, particularly the integration of artificial intelligence and real-time processing systems to improve scalability and performance. To address these challenges, the paper recommends adopting advanced distributed frameworks, strengthening data governance, implementing real-time analytics, and investing in data engineering skills. These strategies aim to improve the efficiency, reliability, and scalability of big data systems, enabling organizations to remain competitive in a data-driven environment.

INTRODUCTION

The increasing amount of data in which digital sources like social media, sensors, and enterprise systems are generated has become an essential part of data engineering work through its requirements for processing big data. Big data refers to data sets which the existing data management tools are unable to handle due to their size, complexity, and speed of change exceeding those tool's capacities (Badman & Kosinski, 2024). Big data uses five characteristics which are defined as "V" elements to assess its performance which includes volume, velocity, variety, veracity, and value (Kulkarni, 2024). Data engineering also focuses on the design, development, and management of the systems that enable efficient data collection, storage, and data processing. Organizations are encouraged to use big data processing methods to gain valuable insights which provide them with data-driven business decisions. They usually need effective solutions for processing big data as they depend on real-time analytics and extensive data integrations to operate their business nowadays. The organizations may face several obstacles when they try to process enormous datasets as they deal with different datatypes and need as many resources to perform accurate computations but their systems are unable to expand their processing capacity. In addition, the demand for distributed computing frameworks and cloud-based technologies increased as the traditional data processing systems could not meet their operational needs. New technologies like parallel-processing and distributed-storage systems are being developed to make it easier, quicker, and more dependable to process large amounts of information (Vinci, 2024). This paper will examine the current methods that are being used, and recommend ways to increase the effectiveness and efficiency of big data systems.

LITERATURE REVIEW

Researchers have studied big data processing for the last ten years as they need methods that can handle huge and complicated datasets which occur across different sectors such as healthcare, finance, manufacturing, and telecommunications. The research review which studied 15 years of academic work shows that big data analytics has developed from data mining techniques into sophisticated systems which provide enhanced performance and greater scalable analysis capacity (Tosi et al., 2024). The existing processing methods and governance frameworks need continuous development as organizations face persistent problems with data quality, system scalability, and protection of their confidential information (Gusman et al., 2022). The research demonstrates that traditional processing systems cannot handle three main characteristics of big data which includes its high volume, rapid processing speed, and diverse data types (Gusman et al, 2022).

The study of big data technology evolution shows that organizations now use artificial intelligence together with real-time processing systems to achieve better decision-making and prediction results which helps them maintain customer trust while meeting big data systems data security requirements (Hakami et al, 2025). The creation of efficient big data processing systems encounters major obstacles as existing research shows that three key areas including data integrity protection, mixed format data processing, and different workload performance optimization need to be further investigated by researchers to integrate scalable and intelligent systems that comply with governance requirements.

ANALYSIS AND DISCUSSION

Organizing and cleaning vast amounts of data of multiple structures such as structured, semi-structured and unstructured data can be referred to as big data processing. There are five stages in big data processing which are data extraction, data transformation, data loading, data visualization and machine learning application. Firstly, structured and unstructured data is gathered from various sources. They are merged, filtered and verified by removing inaccuracies and filling in missing information. Next, converting data into required formats takes part in data transformation. The data is transformed using aggregation, normalisation, feature selection, binning and clustering. Thirdly, data loading is where data is loaded into a centralized database system. These first three processes define the Extract, Transform, Load (ETL) processes. Then, the data is visualized using tools such as Power BI to create dashboards which provide an overview of the organizations. Lastly, machine learning applications focus on developing models that use learning algorithms to rapidly analyze vast amounts of data and adapt to new input.

Use cases of big data processing in real life are in e-commerce and retail and manufacturing. In e-commerce and retail, organizations or companies like Shopee and Lazada use big data processing to identify and recommend products that you might like based on your browsing and purchase history.

Whereas in manufacturing, manufacturers use big data processing to predict and ensure the efficiency of their machines to prevent breakdowns.

Big data processing is important as it enables organizations to identify patterns and derive insights from large datasets. Organizations or businesses can gain insights of customer behaviour and market trends. This enables the organizations to make better, informed decisions. Moreover, organizations can also cut costs by using big data processing. Big data processing allows the organizations to identify inefficiencies, reduce waste, and optimize resource allocation, which ultimately leads to lower operational costs. Big data processing also improves risk management of an organization. This is because big data processing can be used for forecasting. For example, predictive models can assist in identifying possible fraud in the financial industry, whereas in the medical field, they can forecast disease outbreaks. Apart from that, big data processing also gives competitive advantage for organizations. Organizations will be able to identify new opportunities and also respond quickly to market changes.

Challenges in big data processing include volume, velocity, variety and veracity of data. A large amount of data needs to be processed at high speed in a short period of time. The data also comes in different forms such as text, numbers, images, videos, and more and all the data is not reliable, therefore requires the organizations to filter and clean the data. Other than that, big data processing is costly as storing large amounts of data can be expensive. Maintaining privacy and security also becomes an issue in big data processing as it might contain sensitive information. Hence, protecting the data from being breached by hackers is a major concern.

RECOMMENDATIONS

According to Badman and Kosinski (2024), and Gusman et al., (2022), organizations should use advanced processing frameworks which can handle high-speed, high volume, and multiple data types as traditional systems fail to meet current data environment needs according to data engineering requirements for big data processing. The use of artificial intelligence (AI) in big data processing pipelines should be implemented as AI-based methods help the organizations to achieve better results through automated data analysis and decision-making while providing advanced insights from complicated data sets, which will lead to improved system performance (Kumar et al., 2024). Organizations also recommended establishing real-time data processing and analytics systems as these technologies enable quicker decision-making,which leads to better operational performance and business outcomes (Badman & Kosinski, 2024). The process of data governance requires organizations to develop stronger mechanisms for managing data quality, data security, and data privacy protection, which results in trustworthiness when they handle diverse data sources that contain high-volume information (Gusman et al., 2022). The organizations should develop flexible systems which combine both batch and streaming processing to manage their diverse workload requirements. Last but not least, continuous investment in data engineering skills enable the teams in an organization to master emerging big data technologies, which will help the organization stay competitive while using big data for innovative business development and decision-making processes (Badman & Kosinski, 2024).

CONCLUSION

In conclusion, big data processing has become a fundamental component of modern data engineering, allowing organizations to extract valuable insights from big and complex datasets. While traditional systems are no longer sufficient to handle the scale and diversity of big data, emerging technologies such as distributed computing, artificial intelligence, and real-time analytics offer effective solutions. Although big data processing has its advantages, it also has several challenges, including data management complexity, high infrastructure costs, and concerns related to data privacy and security. By adopting scalable and intelligent data processing systems, organizations can enhance decision-making, improve operational efficiency, and gain a competitive advantage. Ultimately, the successful implementation of big data processing strategies will play a critical role in supporting sustainable growth in the digital era.

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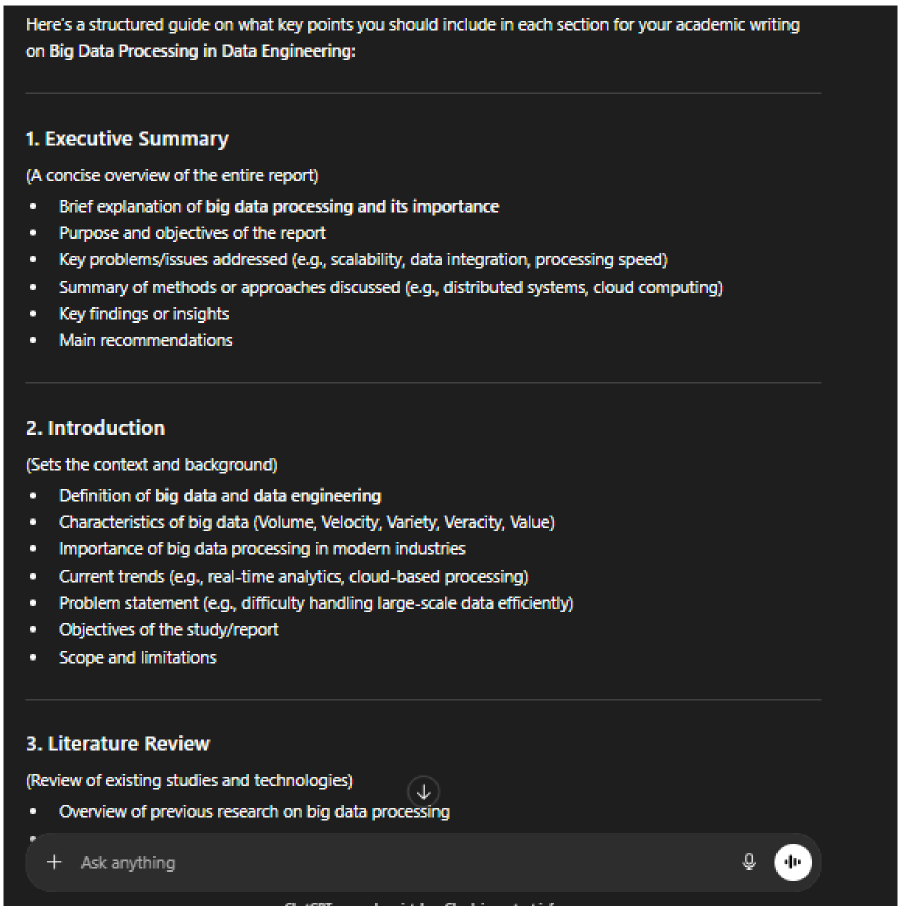
APPENDIX / LOGBOOK


Here's a structured guide on what key points you should include in each section for your academic writing on **Big Data Processing in Data Engineering:**

1. Executive Summary

(A concise overview of the entire report)

- Brief explanation of **big data processing and its importance**
- Purpose and objectives of the report
- Key problems/issues addressed (e.g., scalability, data integration, processing speed)
- Summary of methods or approaches discussed (e.g., distributed systems, cloud computing)
- Key findings or insights
- Main recommendations

2. Introduction

(Sets the context and background)

- Definition of **big data** and **data engineering**
- Characteristics of big data (Volume, Velocity, Variety, Veracity, Value)
- Importance of big data processing in modern industries
- Current trends (e.g., real-time analytics, cloud-based processing)
- Problem statement (e.g., difficulty handling large-scale data efficiently)
- Objectives of the study/report
- Scope and limitations

3. Literature Review

(Review of existing studies and technologies)

- Overview of previous research on big data processing

Ask anything

Date and time used: 3 April 2026, 12:53 PM
Author: Nabil Aflah
Prompt: Provide me points that I should have in my academic writing for topic: Big data processing in data engineering 1. Executive summary 2. Introduction 3. Literature review 4. Analysis and discussion 5. Recommendations 6. Conclusion

Date and time used: 10 April 2026, 3:28 PM
Author: Harini Sangaran
Prompt in Google: Big data processing

Date	Task	Findings	Author
2 April 2026	- Divide tasks equally among team members - Do research on the topic: Big Data Processing in Data Engineering on Scopus	- Read some of the articles that related to the topic	Nabil Aflah
3 April 2026	- Prompt for ideas to write for the report - Prepared materials for introduction	- Write on the introduction part	Nabil Aflah
8 April 2026	- Take a look on the articles that related for literature review	- Write on the literature review	Nabil Aflah
9 April 2026	- From the articles that have gained, prepared for recommendations - Refine the logbook and check for all parts that I was assigned to	- Write on the recommendations part	Nabil Aflah
10 April 2026	- Do research on the topic: Big Data Processing in Data Engineering - Prepared materials for report	- Write on the executive summary, analysis and discussion and conclusion	Harini